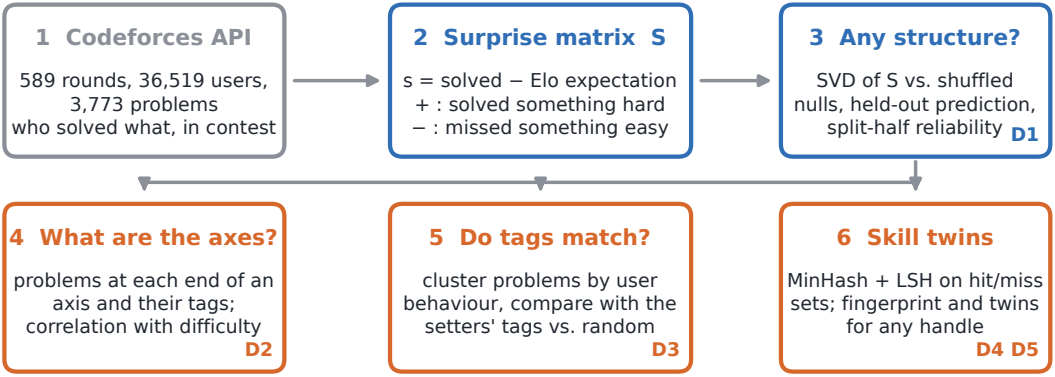
1 Codeforces API
589 rounds, 36,519 users,
3,773 problems
who solved what, in contest
2 Surprise matrix S
s = solved − Elo expectation
+ : solved something hard
− : missed something easy
3 Any structure?
SVD of S vs. shuffled
nulls, held-out prediction,
split-half reliability
D1
4 What are the axes?
problems at each end of an
axis and their tags;
correlation with difficulty
D2
5 Do tags match?
cluster problems by user
behaviour, compare with the
setters' tags vs. random
D3
6 Skill twins
MinHash + LSH on hit/miss
sets; fingerprint and twins
for any handle
D4 D5